Problem Set 2

Applied Stats/Quant Methods 1

Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```
#Question1a
matrix <- matrix(c(14, 6, 7,
                   7, 7, 1),
                 nrow = 2, byrow = TRUE)
rownames(matrix) <- c("Upper", "Lower")
colnames(matrix) <- c("Not_Stopped", "Bribe_Requested", "Stopped_Warning"
    )
matrix
row_totals <- rowSums(matrix)
col_totals <- colSums(matrix)
grand_total <- sum(matrix)
expected <- outer(row_totals, col_totals) / grand_total
expected
chi_sq_stat <- sum((matrix - expected)^2 / expected)
chi_sq_stat
```

The Chi-Squared Statistic obtained is 3.79

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

```
#Question1b
df = 2
p_value <- 1 - pchisq(chi_sq_stat, df)
p_value
```

The P-Value calculated is 0.15, this is greater than $\alpha = 0.1$. Therefore, we fail to reject the null hypothesis. This means that the data does not provide sufficient evidence that at $\alpha = 0.1$ the distribution of outcomes differs by driver class.

[2]Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.136	-0.815	0.819
Lower class	-0.183	1.094	-1.099

```
#Question1c
#Standardized Residuals
chi.test <- chisq.test(matrix)
residuals <- chi.test$residuals
residuals
```

Using the chi-squared test on the contingency table (matrix) and then extracting the residuals from the results, gives the values in the table.

(d) How might the standardized residuals help you interpret the results?

Interpretation:

- Lower-class drivers were slightly more likely to have a bribe requested than expected (r = +1.09).

- They were slightly less likely to be stopped/given a warning (r = −1.10). Differences for upper-class drivers are small and mostly opposite in sign (as expected, since the totals balance).

- Although, none of the residuals exceeded ±2 but we can infer that on average, police may be somewhat more likely to solicit bribes from lower-class drivers and somewhat less likely to just warn them. However, these results are still not statistically significant as p = 0.15 and r<2.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

[3] Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica.* 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

Null Hypothesis: The reservation policy has no effect on the number of new or repaired drinking water facilities. $\beta = 0$

Alternative Hypothesis: The reservation policy has a significant effect (could be positive or negative) on the number of new or repaired drinking-water facilities $\beta \neq 0$

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

```
#Question2b
library(stargazer)
women <- read.csv("women.csv")
model <- lm(water ~ reserved, data = women)
stargazer(model,
          type = "latex",    #
          title = "Effect of Reservation Policy on Drinking-Water Facilities",
          dep.var.labels = "Number of New/Repaired Water Facilities",
          covariate.labels = c("Reserved for Women"),
          table.placement = "!htbp",
          digits = 3,
          align = TRUE,
          no.space = TRUE)
```

Table 1: Effect of Reservation Policy on Drinking-Water Facilities

	Dependent variable:
	Number of New/Repaired Water Facilities
Reserved for Women	9.252**
	(3.948)
Constant	14.738***
	(2.286)
Observations	322
R^2	0.017
Adjusted R^2	0.014
Residual Std. Error	33.446 (df = 320)
F Statistic	5.493** (df = 1; 320)
Note:	*p<0.1; **p<0.05; ***p<0.01

(c) Interpret the coefficient estimate for reservation policy.

The coefficient estimate for the reservation policy is 9.26. Therefore, on average, villages in Gram Panchayats reserved for women leaders have 9.26 more new or repaired drinking-water facilities than villages in non-reserved GPs. This difference is statistically significant ($p < 0.01$), suggesting that the reservation policy increased investments in water infrastructure, consistent with the hypothesis that female leaders prioritize issues women care about, such as drinking water.